

Name: Prathamesh Arvind Jadhav
Roll No: 4059

Experiment No:4

Experiment Title:

Perform Text Pre-processing (Tokenization, Stop Word Removal, and Stemming) of a Paragraph.

Aim:

To perform text pre-processing operations including Tokenization, Stop Word Removal, and Stemming using Python.

Objective:

- Understand the importance of text pre-processing in NLP.
 - Perform sentence and word tokenization using NLTK.
 - Remove stop words from the tokenized text.
 - Apply stemming to reduce words to their root form.
 - Count and visualize stop words and stemmed words.
-

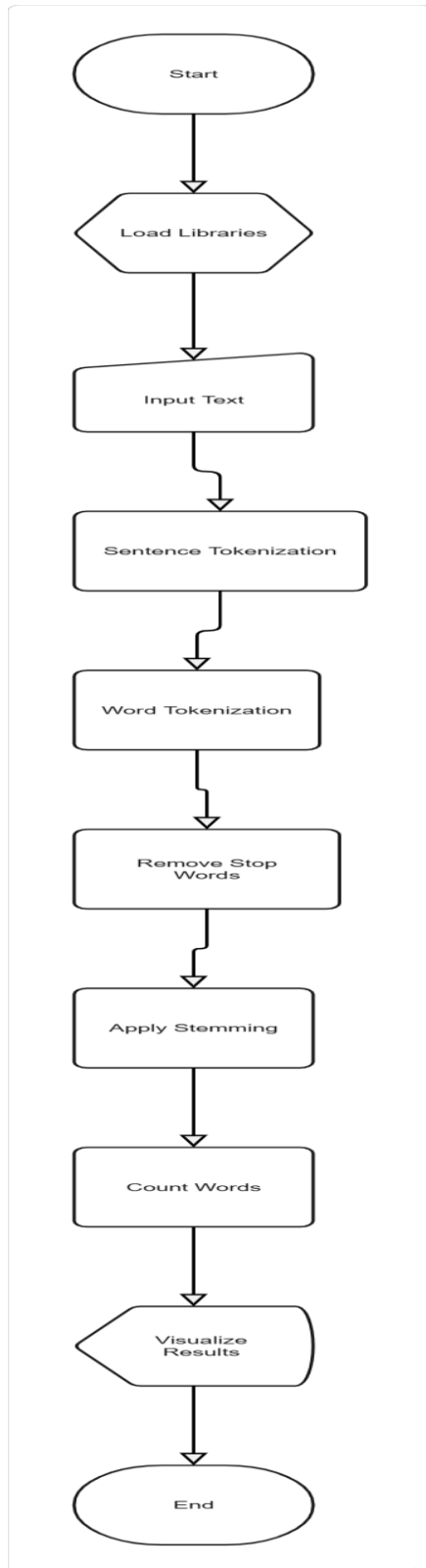
Procedure & Flowchart:

- **Procedure:**

Procedure (Using spaCy):

1. Install spaCy: pip install spacy
2. Define a sample paragraph related to a real-world domain (e.g., health).
3. Perform **sentence tokenization** and **word tokenization**.
4. Remove stop words using NLTK's stopword list.
5. Apply **PorterStemmer** to get the stemmed words.
6. Count the number of stop words and stemmed words.
7. Visualize the counts using a **bar chart**.

- **Flowchart:**



Code / Implementation:

The screenshot shows a Jupyter Notebook titled "NLP_EXP-4.ipynb". The code in cell [2] performs the following steps:

- Imports `nlTK`, `sent_tokenize`, `word_tokenize`, `stopwords`, and `PorterStemmer`.
- Downloads necessary NLTK data: `punkt`, `stopwords`, and `punkt_tab`.
- Defines a sample text: "I am Prathamesh Jadhav, a passionate and dedicated student of Artificial Intelligence and Data Science Engineering with a strong interest in solving real-world problems."
- Tokenizes the text into sentences and words.
- Removes stop words from the list of words.
- Stems the remaining words using `PorterStemmer`.

The output of the code is as follows:

```
Sentences: ['I am Prathamesh Jadhav, a passionate and dedicated student of Artificial Intelligence and Data Science Engineering with a strong interest in solving real-world problems.']
Words: ['I', 'am', 'Prathamesh', 'Jadhav', ',', 'a', 'passionate', 'and', 'dedicated', 'student', 'of', 'Artificial', 'Intelligence', 'and', 'Data', 'Science', 'Engineering', 'with', 'a', 'strong', 'interest', 'in', 'solving', 'real-world', 'problems']
Filtered Words: ['Prathamesh', 'Jadhav', ',', 'passionate', 'dedicated', 'student', 'Artificial', 'Intelligence', 'Data', 'Science', 'Engineering', 'strong', 'interest', 'solving', 'real-world', 'problems']
Stemmed Words: ['prathamesh', 'jadhav', ',', 'passion', 'dedic', 'student', 'artifici', 'intellig', 'data', 'scienc', 'engin', 'strong', 'interest', 'solv', 'real-world', 'problem']
```

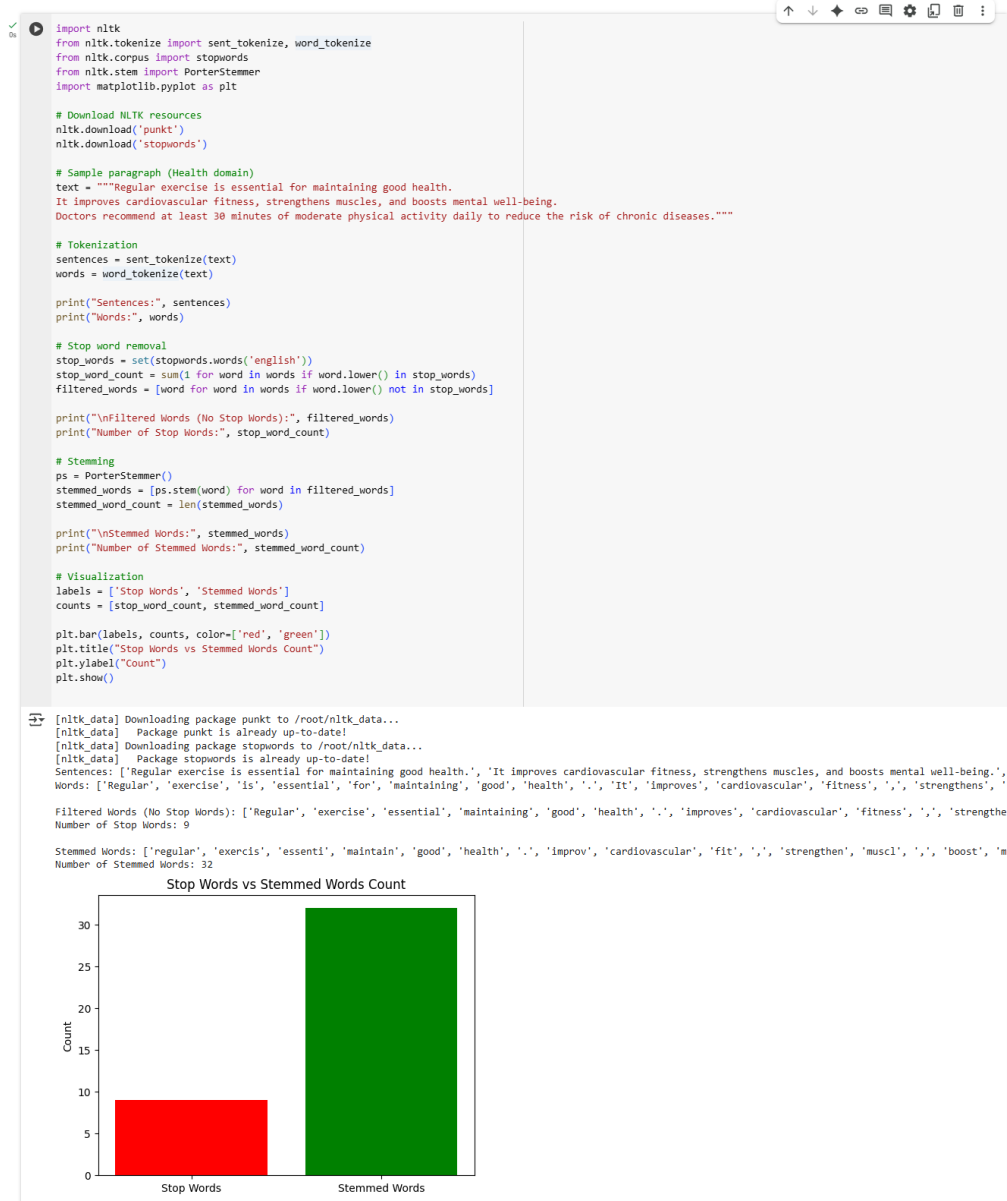
Student Activity - Code & Output

Student Task:

- Take a paragraph from any real-world domain (sports, education, health, etc.).
- Apply **tokenization, stop word removal, and stemming**.
- Count the number of stop words and stemmed words.
- Display the counts using a **bar chart**.

Sample Code for Student Activity:

Student Activity:



Questions & Answers:

Q1. What is the importance of text pre-processing in NLP? (CO1)

Answer:

Text pre-processing cleans and normalizes raw text, making it suitable for analysis. It removes noise, reduces dimensionality, and improves the accuracy of NLP models.

Q2. Differentiate between stemming and lemmatization. (CO1)

Answer:

- **Stemming:** Cuts off word endings without understanding context, e.g., "running" → "run".
- **Lemmatization:** Uses linguistic knowledge to return base word, e.g., "better" → "good".

Q3. How does Porter Stemmer work? (CO3)

Answer:

The Porter Stemmer applies a series of rules to strip suffixes from words in a step-by-step process, reducing them to their stem/root form.

Q4. Why is visualization useful in text processing? (CO5)

Answer:

Visualization helps in quickly understanding word distribution, stop word frequency, and processed text statistics, aiding better analysis.

Conclusion:

In this experiment, we successfully performed **tokenization, stop word removal, and stemming** on a sample text using Python's NLTK library. We also counted stop words and stemmed words, then visualized the results using a bar chart. This process is crucial for preparing clean and meaningful textual data for NLP tasks.